

CO1 - Assessment Tool - 01

1A) Given that

Box A: 6 red, 4 Blue, 2 Green $\rightarrow$ Total = 12

Box B: 5 red, 7 Blue, 3 Green $\rightarrow$ Total = 15

Box C: 8 red, 2 Blue, 5 Green $\rightarrow$ Total = 15

Since B is twice as likely as A & C

$$P(A) = 1/4, P(B) = 1/2, P(C) = 1/4$$

Probability of blue

$$P(\text{Blue}/A) = 4/12 = 1/3$$

$$P(\text{Blue}/B) = 7/15$$

$$P(\text{Blue}/C) = 2/15$$

Using Bayes theorem:-

$$P(C/\text{Blue}) = \frac{P(C) \cdot P(\text{Blue}/C)}{P(A) \cdot P(\text{Blue}/A) + P(B) \cdot P(\text{Blue}/B) + P(C) \cdot P(\text{Blue}/C)}$$

$$= \frac{1/4 \times 2/15}{1/4 \cdot 1/3 + 1/2 \cdot 7/15 + 1/4 \cdot 2/15}$$

$$= \frac{2}{21}$$

$$P(C/\text{Blue}) = 0.0952$$

2A) Given that

Section A: 5 red, 3 Blue, 2 Black $\rightarrow$ Total = 10

Section B: 4 red, 6 Blue, 5 Black $\rightarrow$ Total = 15

Section C: 7 red, 2 Blue, 6 Black $\rightarrow$ Total = 15

B is twice as likely:-

$$P(A) = 1/4, P(B) = 1/2, P(C) = 1/4$$

Blue probabilities:-

$$P(\text{Blue}/A) = 3/10$$

$$P(\text{Blue}/B) = 2/5$$

$$P(\text{Blue}/C) = 2/15$$

$$\therefore P(C/\text{Blue}) = \frac{1/4 \cdot 2/15}{1/4 \cdot 3/10 + 1/2 \cdot 2/5 + 1/4 \cdot 2/15}$$

$$= 4/37$$

$$P(C/\text{Blue}) = 0.1081$$

3A) Given that

The test set contains:-

$$1800 + 1800 = 3600$$

20% is used for testing:-

$$3600 \times 0.2 = 720$$

Each class has

$$1800 \times 0.2 = 360$$

Given:-

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

$$\therefore FN = 360 - 210 = 150$$

Confusion matrix:-

	Actual Positive	Actual Negative
Predictive Positive	TP = 210	FP = 30
Predictive Negative	FN = 150	TN = 190

Accuracy:-

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720}$$

$$= 55.56\%$$

Precision :-

$$= \frac{TP}{TP+FP} = \frac{210}{210+30} = 87.5\%$$

Sensitivity / Recall :-

$$= \frac{TP}{TP+FN} = \frac{210}{210+150} = 58.33\%$$

Specificity :-

$$= \frac{TN}{TN+FP}$$

$$= \frac{190}{190+30} = 86.36\%$$

7A) Given that

$$TP = 119$$

$$FN = 6$$

$$TN = 120$$

$$FP = 5$$

Accuracy :-

$$= \frac{TP+TN}{Total}$$

$$= \frac{119+120}{250} = 95.6\%$$

Precision :-

$$= \frac{TP}{TP+FP} = \frac{119}{119+5} = 95.97\%$$

Sensitivity :-

$$= \frac{TP}{TP+FN} = \frac{120}{120+5} = 96\%$$

F1-Score :-

$$= \frac{2TP}{2TP+FP+FN}$$

$$= \frac{238}{238+5+6} = 95.58\%$$

8A) Given

$$TP = 138$$

$$FN = 12$$

$$TN = 135$$

$$FP = 15$$

$$Total = 300$$

Accuracy :-

$$= \frac{TP+TN}{Total} = \frac{138+135}{300}$$

$$= 91\%$$

Precision :-

$$= \frac{TP}{TP+FP} = \frac{138}{138+15}$$

$$= 90.2\%$$

Sensitivity :-

$$= \frac{TP}{TP+FN} = \frac{135}{135+12} = 92\%$$

Specificity :-

$$= \frac{TN}{TN+FP} = \frac{135}{135+15} = 90\%$$

F1-Score :-

$$= \frac{2TP}{2TP+FP+FN}$$

$$= \frac{2(138)}{2(138)+15+12} = 91.09\%$$